IQRA UNIVERSITY

DEPARTMENT OF COMPUTER AND TECHNOLOGY

DIGITAL LOGIC DESIGN LAB

Lab Manual 1

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Registration Number: 052-26-128036

Date: 11/02/2026

Program: BS Artificial Intelligence

Semester: 2nd

ALL LOGIC GATES

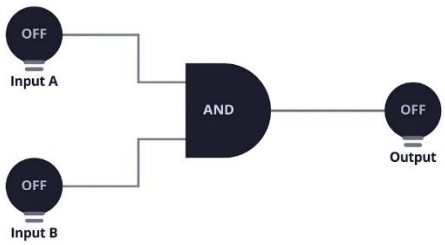
AND GATE

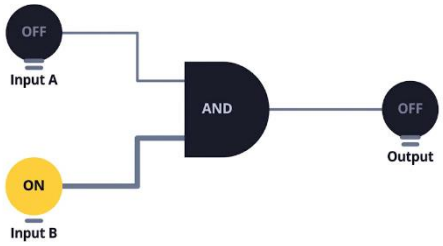
IC Number: 7408

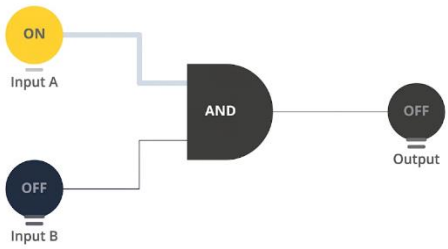
- AND gate gives output 1 only when both inputs are 1. Otherwise, output is 0.
- Boolean Expression: $Y = A \cdot B$

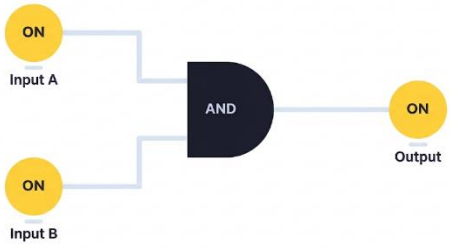
GATE DIAGRAMS











TRUTH TABLE

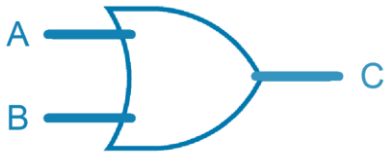
A	B	OUTPUT
0	0	0
0	1	0
1	0	0
1	1	1

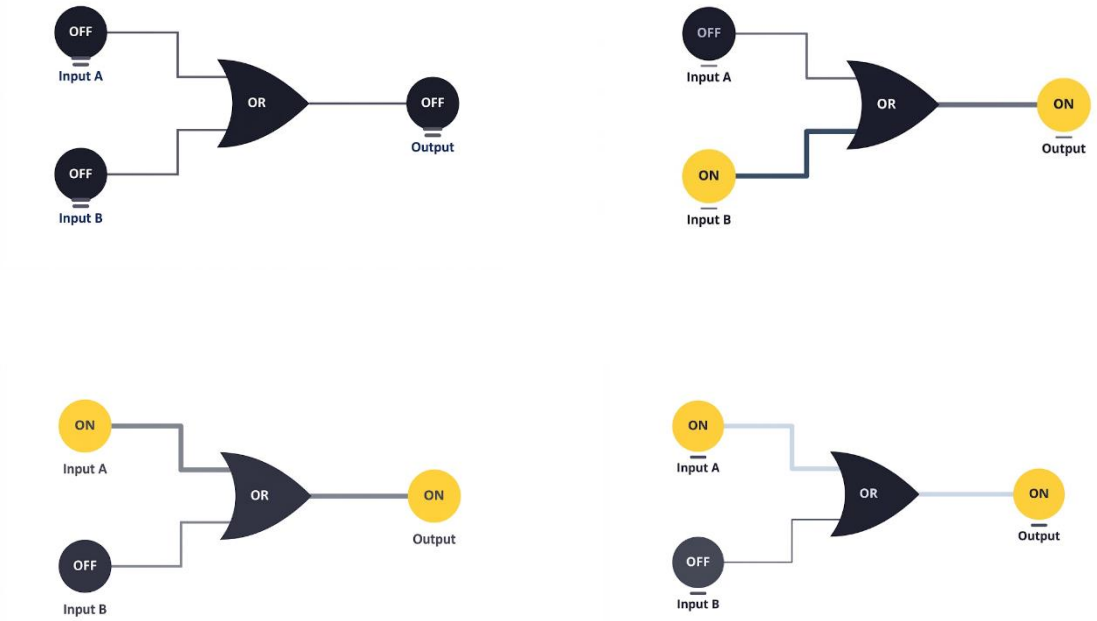
OR GATE

IC Number: 7432

- OR gate gives output 1 when at least one input is 1. Output is 0 only when both inputs are 0.
- Boolean Expression: $Y = A + B$

GATE DIAGRAMS





TRUTH TABLE

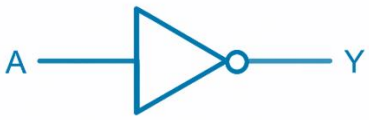
A	B	OUTPUT
0	0	0
0	1	1
1	0	1
1	1	1

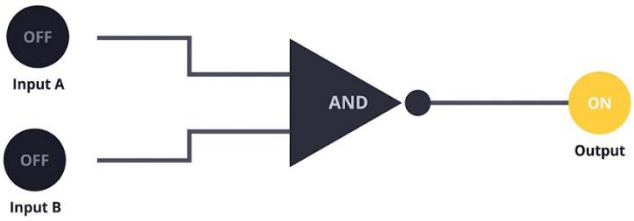
NOT GATE

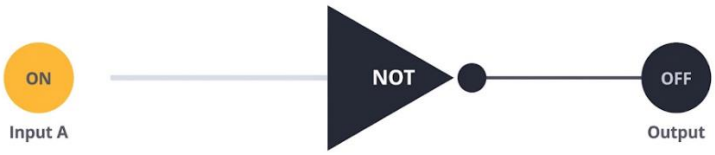
IC Number: 7404

- NOT gate reverses the input. If input is 0, output is 1, and vice versa.
- Boolean Expression: $Y = \overline{A}$

GATE DIAGRAMS







TRUTH TABLE

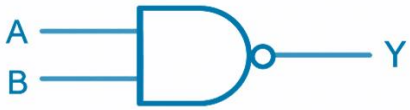
A	OUTPUT
0	1
1	0

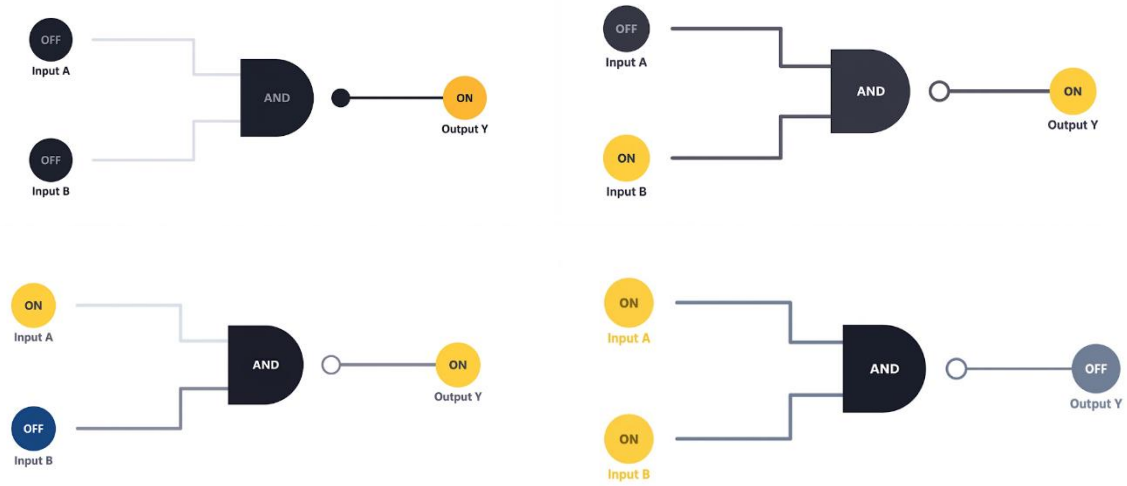
NAND GATE

IC Number: 7400

- Description: NAND gate is the opposite of AND gate. It gives output 0 only when both inputs are 1.
- Boolean Expression: $Y = \overline{A \cdot B}$

GATE DIAGRAMS





TRUTH TABLE

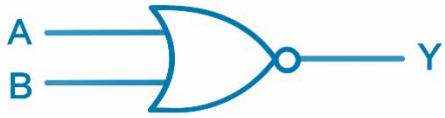
A	B	OUTPUT
0	0	1
0	1	1
1	0	1
1	1	0

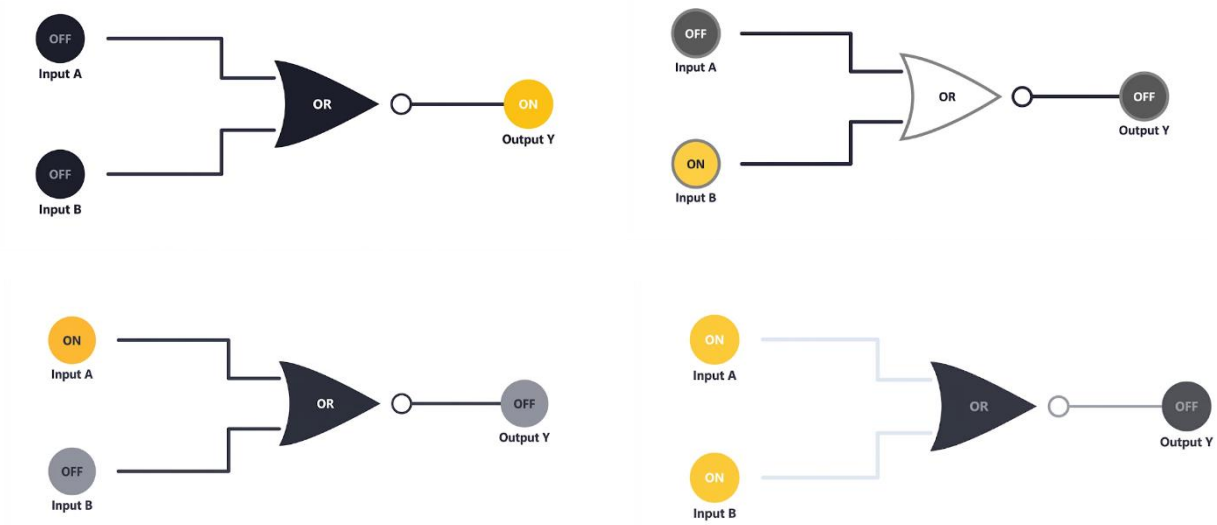
NOR GATE

IC Number: 7402

- NOR gate is the opposite of OR gate. It gives output 1 only when both inputs are 0.
- Boolean Expression: $Y = \overline{A + B}$

GATE DIAGRAMS





TRUTH TABLE

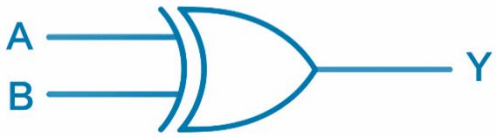
A	B	OUTPUT
0	0	1
0	1	0
1	0	0
1	1	0

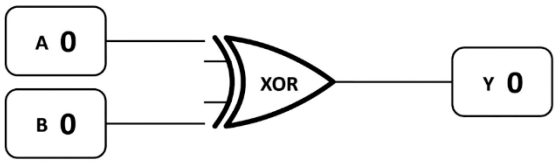
XOR GATE

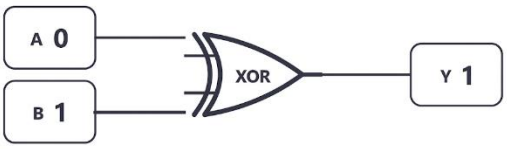
IC Number: 7486

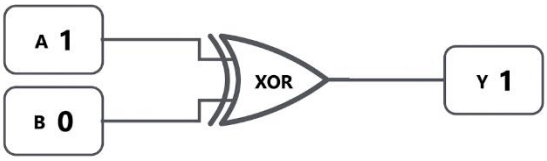
- XOR gate gives output 1 when both inputs are different. If both inputs are the same, output is 0.
- Boolean Expression:
  - $Y = \overline{A}B + A\overline{B}$
  - $Y = A \oplus B$

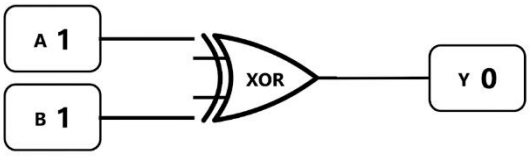
GATE DIAGRAMS











TRUTH TABLE

A	B	Y = XOR
0	0	0
0	1	1
1	0	1
1	1	0